Universität Augsburg
Mathematisch-Naturwissenschaftlich-Technische Fakultät

Classical networks for the Hubbard model with a tilted potential

- Master Thesis Colloquium -

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Chair for theoretical Physics III

12th of March 2025

Outline

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Particle Types

Introduction of the Physical Problem

- Bosonic operators

$$\left[\hat{b}_l, \hat{b}_m\right] = 0$$

$$\left[\hat{b}_l^\dagger, \hat{b}_m^\dagger\right] = 0$$

$$\left[\hat{b}_l, \hat{b}_m^\dagger\right] = \delta_{l,m}$$

Particle Types

Introduction of the Physical Problem

- Bosonic operators
  - Commutation relations

$$\left[\hat{b}_l, \hat{b}_m\right] = 0$$

$$\left[\hat{b}_l^\dagger, \hat{b}_m^\dagger\right] = 0$$

$$\left[\hat{b}_l, \hat{b}_m^\dagger\right] = \delta_{l,m}$$

Particle Types

Introduction of the Physical Problem

- Bosonic operators
  - Commutation relations
- Fermionic operators

$$\left\{ \hat{c}_l, \hat{c}_m \right\} = 0$$

$$\left\{ \hat{c}_l^\dagger, \hat{c}_m^\dagger \right\} = 0$$

$$\left\{ \hat{c}_l, \hat{c}_m^\dagger \right\} = \delta_{l,m}$$

Particle Types

Introduction of the Physical Problem

- Bosonic operators
  - Commutation relations
- Fermionic operators
  - Anti-commutation relations

$$\left\{\hat{c}_l, \hat{c}_m\right\} = 0$$

$$\left\{\hat{c}_l^\dagger, \hat{c}_m^\dagger\right\} = 0$$

$$\left\{\hat{c}_l, \hat{c}_m^\dagger\right\} = \delta_{l,m}$$

Particle Types

Introduction of the Physical Problem

- Bosonic operators
  - Commutation relations
- Fermionic operators
  - Anti-commutation relations
- Hard-core bosonic operators

$$\left[\hat{\mathrm{h}}_l, \hat{\mathrm{h}}_m\right] = 0$$

$$\left[\hat{\mathrm{h}}_l^\dagger, \hat{\mathrm{h}}_m^\dagger\right] = 0$$

$$\left[\hat{\mathrm{h}}_l, \hat{\mathrm{h}}_m^\dagger\right] = \left(1 - 2 \cdot \hat{\mathrm{h}}_m^\dagger \hat{\mathrm{h}}_m\right) \cdot \delta_{l,m}$$

$$\left(\hat{\mathrm{h}}_l^\dagger \hat{\mathrm{h}}_l\right)^n = \hat{\mathrm{h}}_l^\dagger \hat{\mathrm{h}}_l \qquad \forall n \in \mathbb{N}$$

Particle Types

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- Bosonic operators
  - Commutation relations
- Fermionic operators
  - Anti-commutation relations
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Particle Types

Introduction of the Physical Problem

- Bosonic operators
  - Commutation relations
- Fermionic operators
  - Anti-commutation relations
- Hard-core bosonic operators
  - Commutation relations
  - Occupation number limited to 0 and 1

$$\left[\hat{h}_l, \hat{h}_m\right] = 0$$

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Particle Types

Introduction of the Physical Problem

- Bosonic operators
  - Commutation relations
- Fermionic operators
  - Anti-commutation relations
- Hard-core bosonic operators
  - Commutation relations
  - Occupation number limited to 0 and 1
  - Number operator is idempotent

$$\left[\hat{\mathrm{h}}_l, \hat{\mathrm{h}}_m\right] = 0$$

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Particle Types

Introduction of the Physical Problem

- Bosonic operators
  - Commutation relations
- Fermionic operators
  - Anti-commutation relations
- Hard-core bosonic operators
  - Commutation relations
  - Occupation number limited to 0 and 1
  - Number operator is idempotent
- Spins (Pauli operators/matrices)

$$\hat{\sigma}^0 = \mathbb{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\hat{\sigma}^2 = \hat{\sigma}^{\mathrm{y}} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\hat{\sigma}^1 = \hat{\sigma}^{\mathrm{x}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\hat{\sigma}^3 = \hat{\sigma}^{\mathrm{z}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Particle Types

Introduction of the Physical Problem

- Bosonic operators
  - Commutation relations
- Fermionic operators
  - Anti-commutation relations
- Hard-core bosonic operators
  - Commutation relations
  - Occupation number limited to 0 and 1
  - Number operator is idempotent
- Spins (Pauli operators/matrices)
  - Particles from different sites commute
  - Different spin directions on the same site anti-commute
  - They obey angular momentum commutation relations

$$\hat{\sigma}^0 = \mathbb{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\hat{\sigma}^2 = \hat{\sigma}^{\mathrm{y}} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\hat{\sigma}^1 = \hat{\sigma}^{\mathrm{x}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\hat{\sigma}^3 = \hat{\sigma}^{\mathrm{z}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\left\{\hat{\sigma}_l^{\alpha}, \hat{\sigma}_l^{\beta}\right\} = 2 \cdot \delta_{\alpha,\beta}$$

$$\left[\hat{\sigma}_l^{\alpha}, \hat{\sigma}_m^{\beta}\right] = 2 \cdot i \cdot \varepsilon_{\alpha,\beta,\gamma} \cdot \delta_{l,m} \cdot \hat{\sigma}_l^{\gamma}$$

Particle Types - Mappings

Introduction of the Physical Problem

- Properties of Pauli matrices

Particle Types - Mappings

Introduction of the Physical Problem

- Properties of Pauli matrices
  - Base of the complex 2x2 matrices

Particle Types - Mappings

Introduction of the Physical Problem

- Properties of Pauli matrices
  - Base of the complex 2x2 matrices
  - Can be directly mapped to hard-core bosons

$$\hat{\sigma}_l^{x} = \hat{h}_l + \hat{h}_l^{\dagger}$$
$$\hat{\sigma}_l^{y} = i \cdot \left(\hat{h}_l - \hat{h}_l^{\dagger} \right)$$
$$\hat{\sigma}_l^{z} = 2 \cdot \hat{h}_l^{\dagger} \hat{h}_l - 1$$

$$|\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |0\rangle$$

$$\hat{\sigma}^{+} |\downarrow\rangle = |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |1\rangle = \hat{h}^{\dagger} |0\rangle$$

Hamiltonian of the system

Introduction of the Physical Problem

- Hubbard model Hamiltonian

$$\mathcal{H} = \mathcal{H}_0 + \hat{\mathrm{V}}$$

$$\mathcal{H}_0 = U \cdot \sum_{l} \hat{\mathrm{n}}_{l,\uparrow} \hat{\mathrm{n}}_{l,\downarrow} + \sum_{l,\sigma} \underbrace{\left(\vec{E} \cdot \vec{r}_l\right)}_{\varepsilon_l} \hat{\mathrm{n}}_{l,\sigma}$$

$$\hat{\mathrm{V}} = -J \cdot \sum_{\langle l,m \rangle,\sigma} \left(\hat{\mathrm{h}}^{\dagger}_{l,\sigma} \hat{\mathrm{h}}_{m,\sigma} + \hat{\mathrm{h}}^{\dagger}_{m,\sigma} \hat{\mathrm{h}}_{l,\sigma} \right)$$

Hamiltonian of the system

Introduction of the Physical Problem

- Hubbard model Hamiltonian
- Constant external electric field

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Hamiltonian of the system

Introduction of the Physical Problem

- Hubbard model Hamiltonian
- Constant external electric field
- Two particle types (spin degrees)

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$$\hat{\mathrm{V}} = -J \cdot \sum_{\langle l,m \rangle, \sigma} \left(\hat{\mathrm{h}}^{\dagger}_{l,\sigma} \hat{\mathrm{h}}_{m,\sigma} + \hat{\mathrm{h}}^{\dagger}_{m,\sigma} \hat{\mathrm{h}}_{l,\sigma} \right)$$

Hamiltonian of the system

Introduction of the Physical Problem

- Hubbard model Hamiltonian
- Constant external electric field
- Two particle types (spin degrees)
  - Different *spins* than for Pauli spin operators

$$\mathcal{H} = \mathcal{H}_0 + \hat{\mathrm{V}}$$

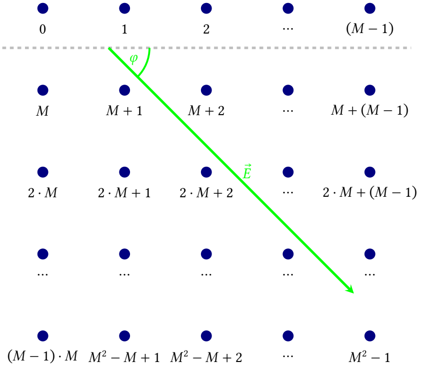
$$\mathcal{H}_0 = U \cdot \sum_{l} \hat{\mathrm{n}}_{l,\uparrow}\hat{\mathrm{n}}_{l,\downarrow} + \sum_{l,\sigma} \underbrace{\left(\vec{E}\cdot\vec{r}_l\right)}_{\varepsilon_l} \hat{\mathrm{n}}_{l,\sigma}$$

$$\hat{\mathrm{V}} = -J \cdot \sum_{\langle l,m\rangle,\sigma} \left(\hat{\mathrm{h}}^{\dagger}_{l,\sigma}\hat{\mathrm{h}}_{m,\sigma} + \hat{\mathrm{h}}^{\dagger}_{m,\sigma}\hat{\mathrm{h}}_{l,\sigma}\right)$$

Geometry of the system

Introduction of the Physical Problem

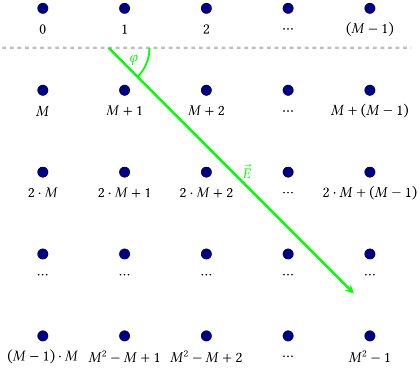
- 2-dimensional geometry



Geometry of the system

Introduction of the Physical Problem

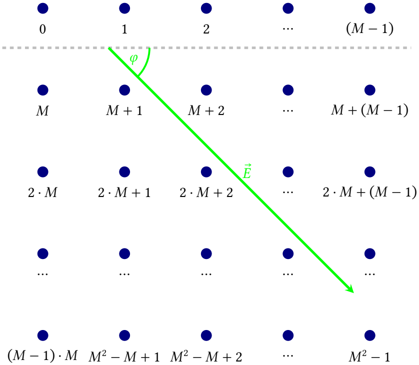
- 2-dimensional geometry
- Square lattice arrangement



Geometry of the system

Introduction of the Physical Problem

- 2-dimensional geometry
- Square lattice arrangement
- Computation for general size and field angle



Goal of the Calculation

Time Evolution

- Evaluate the time-evolution of the system for various observables

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Time Evolution

- Evaluate the time-evolution of the system for various observables
  - Uses calculation in the *Interaction Picture*

Goal of the Calculation

Time Evolution

- Evaluate the time-evolution of the system for various observables
  - Uses calculation in the *Interaction Picture*
  - Introduce operator-less *effective Hamiltonian*

$$\left|\Psi^S(t)\right\rangle = \sum_N e^{\mathcal{H}_{\text{eff}}(N,t)} \Psi_N \left|N\right\rangle$$

The effective Hamiltonian

Time Evolution

$$\mathcal{H}_{\mathrm{eff}}(N,t) = -iE_0(N)t + \mathcal{H}_N(t)$$

The effective Hamiltonian

Time Evolution

$$\mathcal{H}_{\text{eff}}(N,t) = -iE_0(N)t + \mathcal{H}_N(t)$$

- Constructed from the contributions of the base energy and the perturbation

The effective Hamiltonian

Time Evolution

$$\mathcal{H}_{\text{eff}}(N,t) = -iE_0(N)t + \mathcal{H}_N(t)$$

- Constructed from the contributions of the base energy and the perturbation
- Base energy contribution:

$$\begin{aligned} E_0(N) &= \frac{\langle N|\mathcal{H}_0|N\rangle}{\langle N|N\rangle} \\ &= \langle N|U \cdot \sum_l \hat{\mathrm{n}}_{l,\uparrow}\hat{\mathrm{n}}_{l,\downarrow}|N\rangle + \langle N|\sum_{l,\sigma} \varepsilon_l \hat{\mathrm{n}}_{l,\sigma}|N\rangle \\ &= U \cdot \sum_l n_{l,\uparrow}n_{l,\downarrow} + \sum_{l,\sigma} \varepsilon_l n_{l,\sigma} \end{aligned}$$

The effective Hamiltonian

Time Evolution

- To evaluate the time-evolution of the perturbation

$$\hat{\mathrm{V}} = -J \cdot \sum_{\langle l,m \rangle} \left(\hat{\mathrm{h}}_l^\dagger \hat{\mathrm{h}}_m + \hat{\mathrm{h}}_m^\dagger \hat{\mathrm{h}}_l + \hat{\mathrm{d}}_l^\dagger \hat{\mathrm{d}}_m + \hat{\mathrm{d}}_m^\dagger \hat{\mathrm{d}}_l \right)$$

The effective Hamiltonian

Time Evolution

- To evaluate the time-evolution of the perturbation

$$\hat{\mathrm{V}} = -J \cdot \sum_{\langle l,m \rangle} \left(\hat{\mathrm{h}}_l^\dagger \hat{\mathrm{h}}_m + \hat{\mathrm{h}}_m^\dagger \hat{\mathrm{h}}_l + \hat{\mathrm{d}}_l^\dagger \hat{\mathrm{d}}_m + \hat{\mathrm{d}}_m^\dagger \hat{\mathrm{d}}_l \right)$$

- Solve the equation of motion for the ladder operators

$$\hat{\mathrm{h}}_m^{\dagger\mathrm{I}}(t) = e^{i \cdot \varepsilon_m \cdot t} \left(1 + \left(e^{i \cdot U \cdot t} - 1 \right) \hat{\mathrm{d}}_m^{\dagger\mathrm{S}} \hat{\mathrm{d}}_m^{\mathrm{S}} \right) \hat{\mathrm{h}}_m^{\dagger\mathrm{S}}$$

$$\hat{\mathrm{h}}_m^{\mathrm{I}}(t) = e^{-i \cdot \varepsilon_m \cdot t} \left(1 + \left(e^{-i \cdot U \cdot t} - 1 \right) \hat{\mathrm{d}}_m^{\dagger\mathrm{S}} \hat{\mathrm{d}}_m^{\mathrm{S}} \right) \hat{\mathrm{h}}_m^{\mathrm{S}}$$

$$\hat{\mathrm{d}}_m^{\dagger\mathrm{I}}(t) = e^{i \cdot \varepsilon_m \cdot t} \left(1 + \left(e^{i \cdot U \cdot t} - 1 \right) \hat{\mathrm{h}}_m^{\dagger\mathrm{S}} \hat{\mathrm{h}}_m^{\mathrm{S}} \right) \hat{\mathrm{d}}_m^{\dagger\mathrm{S}}$$

$$\hat{\mathrm{d}}_m^{\mathrm{I}}(t) = e^{-i \cdot \varepsilon_m \cdot t} \left(1 + \left(e^{-i \cdot U \cdot t} - 1 \right) \hat{\mathrm{h}}_m^{\dagger\mathrm{S}} \hat{\mathrm{h}}_m^{\mathrm{S}} \right) \hat{\mathrm{d}}_m^{\mathrm{S}}$$

The effective Hamiltonian

Time Evolution

- Insert and reorder the operators

$$
\begin{aligned}
\hat{\mathrm{V}}^{\mathrm{I}}(t) &= \left\{\hat{\mathrm{V}}^{\mathrm{S}}\right\}^{\mathrm{I}}(t) = -J \cdot \sum_{[l,m]} \left\{ \left(\hat{\mathrm{h}}_l^{\dagger\mathrm{S}} \hat{\mathrm{h}}_m^{\mathrm{S}} + \hat{\mathrm{d}}_l^{\dagger\mathrm{S}} \hat{\mathrm{d}}_m^{\mathrm{S}} \right) \right\}^{\mathrm{I}}(t) \\
&= -J \cdot \sum_{[l,m]} \left(\hat{\mathrm{h}}_l^{\dagger\mathrm{I}}(t) \hat{\mathrm{h}}_m^{\mathrm{I}}(t) + \hat{\mathrm{d}}_l^{\dagger\mathrm{I}}(t) \hat{\mathrm{d}}_m^{\mathrm{I}}(t) \right) \\
&= -J \cdot \sum_{[l,m]} \left[\Lambda_{\mathrm{A}}(l,m,t) \cdot \hat{\mathrm{F}}_{\mathrm{A}}(l,m) + \Lambda_{\mathrm{B}}(l,m,t) \cdot \hat{\mathrm{F}}_{\mathrm{B}}(l,m) + \Lambda_{\mathrm{C}}(l,m,t) \cdot \hat{\mathrm{F}}_{\mathrm{C}}(l,m) \right]
\end{aligned}
$$

The effective Hamiltonian

Time Evolution

- Insert and reorder the operators

$$\hat{\mathrm{V}}^{\mathrm{I}}(t) = \left\{\hat{\mathrm{V}}^{\mathrm{S}}\right\}^{\mathrm{I}}(t) = -J \cdot \sum_{[l,m]} \left\{\left(\hat{\mathrm{h}}_l^{\dagger\mathrm{S}}\hat{\mathrm{h}}_m^{\mathrm{S}} + \hat{\mathrm{d}}_l^{\dagger\mathrm{S}}\hat{\mathrm{d}}_m^{\mathrm{S}}\right)\right\}^{\mathrm{I}}(t)$$

$$= -J \cdot \sum_{[l,m]} \left(\hat{\mathrm{h}}_l^{\dagger\mathrm{I}}(t)\hat{\mathrm{h}}_m^{\mathrm{I}}(t) + \hat{\mathrm{d}}_l^{\dagger\mathrm{I}}(t)\hat{\mathrm{d}}_m^{\mathrm{I}}(t)\right)$$

$$= -J \cdot \sum_{[l,m]} \left[\Lambda_{\mathrm{A}}(l,m,t) \cdot \hat{\mathrm{F}}_{\mathrm{A}}(l,m) + \Lambda_{\mathrm{B}}(l,m,t) \cdot \hat{\mathrm{F}}_{\mathrm{B}}(l,m) + \Lambda_{\mathrm{C}}(l,m,t) \cdot \hat{\mathrm{F}}_{\mathrm{C}}(l,m)\right]$$

$$\Lambda_{\mathrm{A}}(l,m,t) = e^{i\cdot(\varepsilon_l-\varepsilon_m)\cdot t} \qquad \hat{\mathrm{F}}_{\mathrm{A}}(l,m) = \sum_{\sigma\in\{\uparrow,\downarrow\}} \hat{\mathrm{h}}_{l,\sigma}^{\dagger\mathrm{S}}\hat{\mathrm{h}}_{m,\sigma}^{\mathrm{S}}\left(1 + 2\cdot\hat{\mathrm{n}}_{l,\overline{\sigma}}^{\mathrm{S}}\hat{\mathrm{n}}_{m,\overline{\sigma}}^{\mathrm{S}} - \hat{\mathrm{n}}_{l,\overline{\sigma}}^{\mathrm{S}} - \hat{\mathrm{n}}_{m,\overline{\sigma}}^{\mathrm{S}}\right)$$

$$\Lambda_{\mathrm{B}}(l,m,t) = e^{i\cdot(\varepsilon_l-\varepsilon_m+U)\cdot t} \qquad \hat{\mathrm{F}}_{\mathrm{B}}(l,m) = \sum_{\sigma\in\{\uparrow,\downarrow\}} \hat{\mathrm{h}}_{l,\sigma}^{\dagger\mathrm{S}}\hat{\mathrm{h}}_{m,\sigma}^{\mathrm{S}}\left(\hat{\mathrm{n}}_{l,\overline{\sigma}}^{\mathrm{S}} - \hat{\mathrm{n}}_{l,\overline{\sigma}}^{\mathrm{S}}\hat{\mathrm{n}}_{m,\overline{\sigma}}^{\mathrm{S}}\right)$$

$$\Lambda_{\mathrm{C}}(l,m,t) = e^{i\cdot(\varepsilon_l-\varepsilon_m-U)\cdot t} \qquad \hat{\mathrm{F}}_{\mathrm{C}}(l,m) = \sum_{\sigma\in\{\uparrow,\downarrow\}} \hat{\mathrm{h}}_{l,\sigma}^{\dagger\mathrm{S}}\hat{\mathrm{h}}_{m,\sigma}^{\mathrm{S}}\left(\hat{\mathrm{n}}_{m,\overline{\sigma}}^{\mathrm{S}} - \hat{\mathrm{n}}_{m,\overline{\sigma}}^{\mathrm{S}}\hat{\mathrm{n}}_{l,\overline{\sigma}}^{\mathrm{S}}\right)$$

The effective Hamiltonian

Time Evolution

- Evaluate the contribution to the effective Hamiltonian

The effective Hamiltonian

Time Evolution

- Evaluate the contribution to the effective Hamiltonian
  - Requires previously calculated value of the V-operator in the Interaction Picture

$$
\begin{aligned}
\mathcal{H}_N(t) &= \sum_{v=1}^{\infty} \frac{(-i)^v}{v!} \int_0^t \mathrm{d}t_1 \int_0^t \mathrm{d}t_2 \cdots \int_0^t \mathrm{d}t_v \left\langle \mathbb{T}\, \hat{\mathrm{V}}^{\mathrm{I}}(t_1)\hat{\mathrm{V}}^{\mathrm{I}}(t_2)\cdots\hat{\mathrm{V}}^{\mathrm{I}}(t_v)\right\rangle_{\mathrm{c}(N)} \\
&= -i \int_0^t \mathrm{d}t_1 \frac{\left\langle N\middle|\hat{\mathrm{V}}^{\mathrm{I}}(t_1)\middle|\Psi^{\mathrm{S}}\right\rangle}{\langle N|\Psi^{\mathrm{S}}\rangle} \\
&- \frac{1}{2}\int_0^t \mathrm{d}t_1 \int_0^t \mathrm{d}t_2 \left(\frac{\left\langle N\middle|\mathbb{T}\hat{\mathrm{V}}^{\mathrm{I}}(t_1)\hat{\mathrm{V}}^{\mathrm{I}}(t_2)\middle|\Psi^{\mathrm{S}}\right\rangle}{\langle N|\Psi^{\mathrm{S}}\rangle} - \frac{\left\langle N\middle|\hat{\mathrm{V}}^{\mathrm{I}}(t_1)\middle|\Psi^{\mathrm{S}}\right\rangle \cdot \left\langle N\middle|\hat{\mathrm{V}}^{\mathrm{I}}(t_2)\middle|\Psi^{\mathrm{S}}\right\rangle}{\langle N|\Psi^{\mathrm{S}}\rangle^2} \right) \\
&+ \cdots
\end{aligned}
$$

The effective Hamiltonian

Time Evolution

- Evaluate the contribution to the effective Hamiltonian
  - Requires previously calculated value of the V-operator in the Interaction Picture
  - Controllable *cumulant expansion*

$$
\begin{aligned}
\mathcal{H}_N(t) &= \sum_{v=1}^{\infty} \frac{(-i)^v}{v!} \int_0^t \mathrm{d}t_1 \int_0^t \mathrm{d}t_2 \cdots \int_0^t \mathrm{d}t_v \left\langle \mathbb{T}\, \hat{\mathrm{V}}^{\mathrm{I}}(t_1) \hat{\mathrm{V}}^{\mathrm{I}}(t_2) \cdots \hat{\mathrm{V}}^{\mathrm{I}}(t_v) \right\rangle_{\mathrm{c}(N)} \\
&= -i \int_0^t \mathrm{d}t_1 \frac{\left\langle N \middle| \hat{\mathrm{V}}^{\mathrm{I}}(t_1) \middle| \Psi^{\mathrm{S}} \right\rangle}{\langle N | \Psi^{\mathrm{S}} \rangle} \\
&- \frac{1}{2} \int_0^t \mathrm{d}t_1 \int_0^t \mathrm{d}t_2 \left(\frac{\left\langle N \middle| \mathbb{T} \hat{\mathrm{V}}^{\mathrm{I}}(t_1) \hat{\mathrm{V}}^{\mathrm{I}}(t_2) \middle| \Psi^{\mathrm{S}} \right\rangle}{\langle N | \Psi^{\mathrm{S}} \rangle} - \frac{\left\langle N \middle| \hat{\mathrm{V}}^{\mathrm{I}}(t_1) \middle| \Psi^{\mathrm{S}} \right\rangle \cdot \left\langle N \middle| \hat{\mathrm{V}}^{\mathrm{I}}(t_2) \middle| \Psi^{\mathrm{S}} \right\rangle}{\langle N | \Psi^{\mathrm{S}} \rangle^2} \right) \\
&+ \cdots
\end{aligned}
$$

Handling of Observables

Time Evolution

- This allows for general evaluation of expectation values

$$\frac{\left\langle \Psi^{S}(t) \middle| \hat{\mathcal{O}} \middle| \Psi^{S}(t) \right\rangle}{\langle \Psi^{S}(t) | \Psi^{S}(t) \rangle} =$$

Handling of Observables

Time Evolution

- This allows for general evaluation of expectation values
  - Requires a probability for the state
  - Requires a local observable

$$\frac{\left\langle \Psi^{S}(t) \middle| \hat{\mathcal{O}} \middle| \Psi^{S}(t) \right\rangle}{\langle \Psi^{S}(t) | \Psi^{S}(t) \rangle} =$$

$$\sum_{N} \mathrm{P}(N,t) \underbrace{\sum_{K} \langle N | \hat{\mathcal{O}} | K \rangle \, e^{\mathcal{H}_{\text{eff}}(K,t) - \mathcal{H}_{\text{eff}}(N,t)} \frac{\Psi_K}{\Psi_N}}_{\hat{\mathcal{O}}_{\text{loc}}(N,t)} = \sum_{N} \mathrm{P}(N,t) \hat{\mathcal{O}}_{\text{loc}}(N,t)$$

Simple Observables

Time Evolution

- Local observable for double occupation measurement

Simple Observables

Time Evolution

- Local observable for double occupation measurement
  - Observables generally are very sparse matrices
  - Reduces to pure occupation measurement

$$\hat{\mathcal{O}}_{\text{loc}}(N,t) = \sum_K \langle N|\hat{\mathcal{O}}_{\text{do}}(l)|K\rangle \, e^{\mathcal{H}_{\text{eff}}(K,t)-\mathcal{H}_{\text{eff}}(N,t)} \frac{\Psi_K}{\Psi_N} = n_{l,\uparrow} \cdot n_{l,\downarrow}$$

Simple Observables

Time Evolution

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- Local observable for particle current measurement

Simple Observables

Time Evolution

- Local observable for double occupation measurement
  - Observables generally are very sparse matrices
  - Reduces to pure occupation measurement

$$\hat{\mathcal{O}}_{\text{loc}}(N,t) = \sum_{K} \langle N|\hat{\mathcal{O}}_{\text{do}}(l)|K\rangle \, e^{\mathcal{H}_{\text{eff}}(K,t)-\mathcal{H}_{\text{eff}}(N,t)} \frac{\Psi_K}{\Psi_N} = n_{l,\uparrow} \cdot n_{l,\downarrow}$$

- Local observable for particle current measurement
  - Requires evaluation of the difference of two effective Hamiltonians

$$\hat{\mathcal{O}}_{\text{loc}}(N,t) = i \cdot \left[n_{m,\sigma} \cdot (1-n_{l,\sigma}) - n_{l,\sigma} \cdot (1-n_{m,\sigma})\right] \cdot \frac{\Psi_{\tilde{N}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N},t)-\mathcal{H}_{\text{eff}}(N,t)}$$

Access to Density-Matrics

Time Evolution

- Non-classical (*quantum*) measurements depend on the (reduced) density matrix

Access to Density-Matrics

Time Evolution

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  - Access to *purity*, *concurrence* and other entanglement measures/monotones

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Time Evolution

- Non-classical (*quantum*) measurements depend on the (reduced) density matrix
  - Access to *purity*, *concurrence* and other entanglement measures/monotones
- Direct calculation not possible

$$\rho_A = \mathrm{Tr}_B\left(\rho_{AB}\right) = \sum_{k=0}^{M} \left(\mathbb{1}_A \otimes \langle \chi_k|\right) \rho_{AB} \left(\mathbb{1}_A \otimes |\chi_k\rangle\right)$$

Access to Density-Matrics

Time Evolution

- Non-classical (*quantum*) measurements depend on the (reduced) density matrix
  - Access to *purity*, *concurrence* and other entanglement measures/monotones
- Direct calculation not possible

$$\rho_A = \mathrm{Tr}_B\left(\rho_{AB}\right) = \sum_{k=0}^{M} \left(\mathbb{1}_A \otimes \langle \chi_k|\right) \rho_{AB} \left(\mathbb{1}_A \otimes |\chi_k\rangle\right)$$

- Use Pauli matrices to expand the complex 4x4 matrix

$$\rho_A(t) = \rho_{l,\sigma,m,\mu}(t) = \frac{1}{4} \sum_{\alpha,\beta\in\{0,\mathrm{x},\mathrm{y},\mathrm{z}\}} \left\langle \hat{\sigma}^{\alpha}_{l,\sigma} \hat{\sigma}^{\beta}_{m,\mu} \right\rangle_{(t)} \left(\hat{\sigma}^{\alpha} \otimes \hat{\sigma}^{\beta}\right)$$

Access to Density-Matrics

Time Evolution

$$\hat{\mathcal{O}} = \hat{\sigma}^{\mathrm{x}}_{l,\sigma}\hat{\sigma}^{0}_{m,\mu} \;:\quad \hat{\mathcal{O}}_{\mathrm{loc}}(N,t) = \frac{\Psi_{\tilde{N}}}{\Psi_N} \cdot e^{\mathcal{H}_{\mathrm{eff}}(\tilde{N},t)-\mathcal{H}_{\mathrm{eff}}(N,t)}$$

$$\hat{\mathcal{O}} = \hat{\sigma}^{\mathrm{y}}_{l,\sigma}\hat{\sigma}^{0}_{m,\mu} \;:\quad \hat{\mathcal{O}}_{\mathrm{loc}}(N,t) = \frac{\Psi_{\tilde{N}}}{\Psi_N} \cdot e^{\mathcal{H}_{\mathrm{eff}}(\tilde{N},t)-\mathcal{H}_{\mathrm{eff}}(N,t)} \cdot i \cdot (1 - 2 \cdot n_{l,\sigma})$$

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$$\hat{\mathcal{O}} = \hat{\sigma}^{\mathrm{x}}_{l,\sigma}\hat{\sigma}^{\mathrm{x}}_{m,\mu} \;:\quad \hat{\mathcal{O}}_{\mathrm{loc}}(N,t) = \frac{\Psi_{\tilde{\tilde{N}}}}{\Psi_N} \cdot e^{\mathcal{H}_{\mathrm{eff}}(\tilde{\tilde{N}},t)-\mathcal{H}_{\mathrm{eff}}(N,t)}$$

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Probability Calculations

Sampling & Simplifications

- Full probability still requires normalization

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- Switch to the Metropolis *Hastings* algorithm
  - Non-exponential number of samples

$$\frac{\left\langle \Psi^{\mathrm{S}}(t) \middle| \hat{\mathcal{O}} \middle| \Psi^{\mathrm{S}}(t) \right\rangle}{\langle \Psi^{\mathrm{S}}(t) | \Psi^{\mathrm{S}}(t) \rangle} = \sum_N \mathrm{P}(N,t) \cdot \hat{\mathcal{O}}_{\mathrm{loc}}(N,t) \approx \frac{1}{|\{N\}_{\mathrm{MC}}|} \sum_{\{N\}_{\mathrm{MC}}} \hat{\mathcal{O}}_{\mathrm{loc}}(N,t)$$

Probability Calculations

Sampling & Simplifications

- Full probability still requires normalization
- Switch to the Metropolis *Hastings* algorithm
  - Non-exponential number of samples
  - Depends only on a transition probability
  - Does not need normalization because it cancels

$$\begin{aligned}
\alpha &= \frac{\mathrm{P}(\tilde{N},t)}{\mathrm{P}(N,t)} = \frac{f(\tilde{N},t)}{f(N,t)} = \frac{\left|e^{\mathcal{H}_{\text{eff}}(\tilde{N},t)}\right|^2 \left|\Psi_{\tilde{N}}\right|^2}{\left|e^{\mathcal{H}_{\text{eff}}(N,t)}\right|^2 \left|\Psi_N\right|^2} \\
&= \frac{\left|\Psi_{\tilde{N}}\right|^2}{\left|\Psi_N\right|^2} \frac{e^{\Re(\mathcal{H}_{\text{eff}}(\tilde{N},t))+i\Im(\mathcal{H}_{\text{eff}}(\tilde{N},t))+\Re(\mathcal{H}_{\text{eff}}(\tilde{N},t))-i\Im(\mathcal{H}_{\text{eff}}(\tilde{N},t))}}{e^{\Re(\mathcal{H}_{\text{eff}}(N,t))+i\Im(\mathcal{H}_{\text{eff}}(N,t))+\Re(\mathcal{H}_{\text{eff}}(N,t))-i\Im(\mathcal{H}_{\text{eff}}(N,t))}} \\
&= \frac{\left|\Psi_{\tilde{N}}\right|^2}{\left|\Psi_N\right|^2} e^{2\cdot\Re(\mathcal{H}_{\text{eff}}(\tilde{N},t))-2\cdot\Re(\mathcal{H}_{\text{eff}}(N,t))} \\
&= \frac{\left|\Psi_{\tilde{N}}\right|^2}{\left|\Psi_N\right|^2} e^{2\cdot\Re(\mathcal{H}_{\text{eff}}(\tilde{N},t)-\mathcal{H}_{\text{eff}}(N,t))}
\end{aligned}$$

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- Choose a helpful initial state

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$$\Psi_N = \frac{1}{\sqrt{\#(\text{states})}} = \frac{1}{\sqrt{2^{\#(\text{sites})\cdot 2}}} = \frac{1}{2^{\#(\text{sites})}}$$

$$\left|\Psi^{\mathrm{S}}(t=0)\right\rangle = \bigotimes_{l=1}^{\#(\text{states})} \frac{1}{2}\left(1 + \hat{\mathrm{h}}_{l,\uparrow}^{\dagger\mathrm{S}} + \hat{\mathrm{h}}_{l,\downarrow}^{\dagger\mathrm{S}} + \hat{\mathrm{h}}_{l,\uparrow}^{\dagger\mathrm{S}}\hat{\mathrm{h}}_{l,\downarrow}^{\dagger\mathrm{S}}\right)|0\rangle$$

- Now *almost all* terms cancel in *differences* of the effective Hamiltonian

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- Now *almost all* terms cancel in *differences* of the effective Hamiltonian
  - Exemplary for single flip on base energy term

$$\begin{aligned} E_0(N) - E_0(\widetilde{N}) &= U \sum_l n_{l,\downarrow} n_{l,\uparrow} - U \sum_l \widetilde{n}_{l,\downarrow}\widetilde{n}_{l,\uparrow} + \sum_{l,\sigma} \varepsilon_l n_{l,\sigma} - \sum_{l,\sigma} \varepsilon_l \widetilde{n}_{l,\sigma} \\ &= \varepsilon_i \left(2n_{i,\sigma_i} - 1\right) + U \cdot \begin{cases} \sigma_i = \uparrow: & n_{i,\downarrow}(2n_{i,\uparrow} - 1) \\ \sigma_i = \downarrow: & n_{i,\uparrow}(2n_{i,\downarrow} - 1) \end{cases} \end{aligned}$$

Why and How should Parameters be variational?

Variational Classical Networks

- Theoretically: always better approximation through higher order terms

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$$O_k(N) = \frac{\partial \ln \Psi_{N,\vec{\eta}}(t)}{\partial \eta_k} = \frac{\partial \ln}{\partial \eta_k} e^{\mathcal{H}_{\text{eff}}(N,\vec{\eta},t)+\ln \Psi_N}$$

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$$\sum_{k'} \overleftrightarrow{S}_{k,k'}\dot{\vec{\eta}}_{k'} = -i\vec{F}_k$$

$$\dot{\vec{\eta}} = -i\overleftrightarrow{S}^{-1}\vec{F}$$

First Try: Cumulant Expansion Prefactors

Variational Classical Networks

- Idea: generate a variational Hamiltonian by replacing the analytical prefactors

$$\mathcal{H}_N(t) = \sum_i C_i(t) \cdot \Phi_i(N) \quad \leftrightarrow \quad \mathcal{H}_{\mathrm{VCN}}(N, \vec{\eta}) = \sum_i \eta_i \cdot \Phi_i(N)$$

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$$C_1(t) = \Pi_{\text{A}}(0,1,t) \qquad \Phi_1(N) = J \sum_l \sum_{\langle l \overset{+(x)}{<} m \rangle} \sum_K \frac{\Psi_K}{\Psi_N} \langle N | \hat{\text{F}}_{\text{A}}(l,m) | K \rangle$$

$$C_2(t) = \Pi_{\text{B}}(0,1,t) \qquad \Phi_2(N) = J \sum_l \sum_{\langle l \overset{+(x)}{<} m \rangle} \sum_K \frac{\Psi_K}{\Psi_N} \langle N | \hat{\text{F}}_{\text{B}}(l,m) | K \rangle$$

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$$C_4(t) = \Pi_{\text{A}}(1,0,t) \qquad \Phi_4(N) = J \sum_l \sum_{\langle l \overset{+(x)}{>} m \rangle} \sum_K \frac{\Psi_K}{\Psi_N} \langle N | \hat{\text{F}}_{\text{A}}(l,m) | K \rangle$$

$$C_5(t) = \Pi_{\text{B}}(1,0,t) \qquad \Phi_5(N) = J \sum_l \sum_{\langle l \overset{+(x)}{>} m \rangle} \sum_K \frac{\Psi_K}{\Psi_N} \langle N | \hat{\text{F}}_{\text{B}}(l,m) | K \rangle$$

$$C_6(t) = \Pi_{\text{C}}(1,0,t) \qquad \Phi_6(N) = J \sum_l \sum_{\langle l \overset{+(x)}{>} m \rangle} \sum_K \frac{\Psi_K}{\Psi_N} \langle N | \hat{\text{F}}_{\text{C}}(l,m) | K \rangle$$

$$C_7(t) = \Pi_{\text{A}}(0,M,t) \qquad \Phi_7(N) = J \sum_l \sum_{\langle l \overset{+(y)}{<} m \rangle} \sum_K \frac{\Psi_K}{\Psi_N} \langle N | \hat{\text{F}}_{\text{A}}(l,m) | K \rangle$$

$$\cdots \qquad \cdots$$

Correction: Watch Explicit Time-Dependency

Variational Classical Networks

- First strategy is not suitable
  - Energy- and variance is not conserved

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- Problem: explicit time-dependency of base energy

$$\frac{\partial \mathcal{H}_{\text{eff}}(N,\vec{\eta},t)}{\partial \vec{\eta}_k} = \frac{\partial\Big(-iE_0(N)t+\mathcal{H}_{\text{VCN}}(N,\vec{\eta})\Big)}{\partial \vec{\eta}_k}$$

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- Solution: Replace base energy factors with variational parameters

$$E_0(N) = U \cdot \sum_m n_{m,\uparrow} n_{m,\downarrow} + \sum_{l,\sigma} \varepsilon_l n_{l,\sigma}$$

$$\eta_l^{\text{b.e.}} = \begin{cases} l=0: & -i \cdot U \cdot t \\ l>0: & -i \cdot \varepsilon_l \cdot t \end{cases}$$

$$\Phi_l^{\text{b.e.}}(N) = \begin{cases} l=0: & \sum_m n_{m,\uparrow} n_{m,\downarrow} \\ l>0: & \sum_\sigma n_{l,\sigma} \end{cases}$$

$$\mathcal{H}_{\text{eff}}^{\text{VCN}}(N,\vec{\eta}) = \sum_{n=1}^{\#(\text{var.})} \eta_n^{\text{var.}} \cdot \Phi_n^{\text{var.}}(N) + \sum_{l=0}^{L} \eta_l^{\text{b.e.}} \cdot \Phi_l^{\text{b.e.}}(N) = \sum_l \eta_l \cdot \Phi_l(N)$$

$$\vec{\eta} = \left(\eta_1^{\text{var.}}, \eta_2^{\text{var.}}, \ldots, \eta_{\#(\text{var.})}^{\text{var.}}, \eta_0^{\text{b.e.}}, \eta_1^{\text{b.e.}}, \ldots, \eta_L^{\text{b.e.}}\right)^{\text{T}}$$

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Implementation & Experiments

- Too much code to properly show
    - Look at repository for overview

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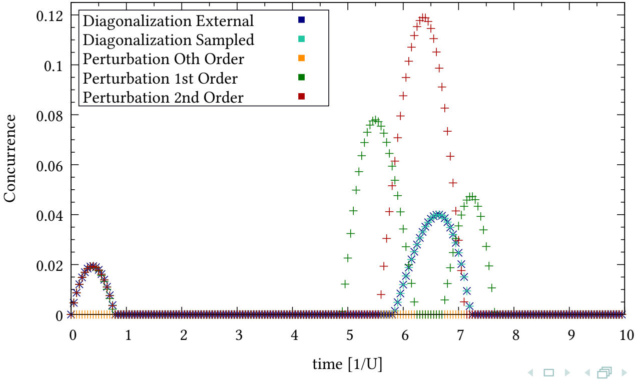
  - Sanity checks for state of data and program
- Optimization: runtime comparison for different code styles
  - For generated code fast execution over readability

Numerical Experiments

Implementation & Experiments

- Sadly no presentation possible in the small time allocated for the presentation
- The plots are in the presentation in the supplementary material



Conclusion & Outlook

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- Successfully transferred and validated the theory
- Working implementation
  - Extensively tested
  - Configurable and extensible
- Started validating expectations with measurements on the compute cluster

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Outlook:

- Re-write in lower level language to gain performance over Python
- Start more real experiments now that the theory is validated to work

Thank you for your kind attention

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Technische Fakultät

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References I

Summary & Conclusion

[1] J. Kell, *Latex sources for this presentation and the main thesis*, (2025) https://github.com/jonas-kell/master-thesis-documents.

[2] J. Kell, *Thesis code implementation and reference*, (2025) https://github.com/jonas-kell/master-thesis-code.

[3] J. Kell, *Https://github.com/jonas-kell/beamer-theme-unia-mntf*, (2024) https://github.com/jonas-kell/math-manipulator.

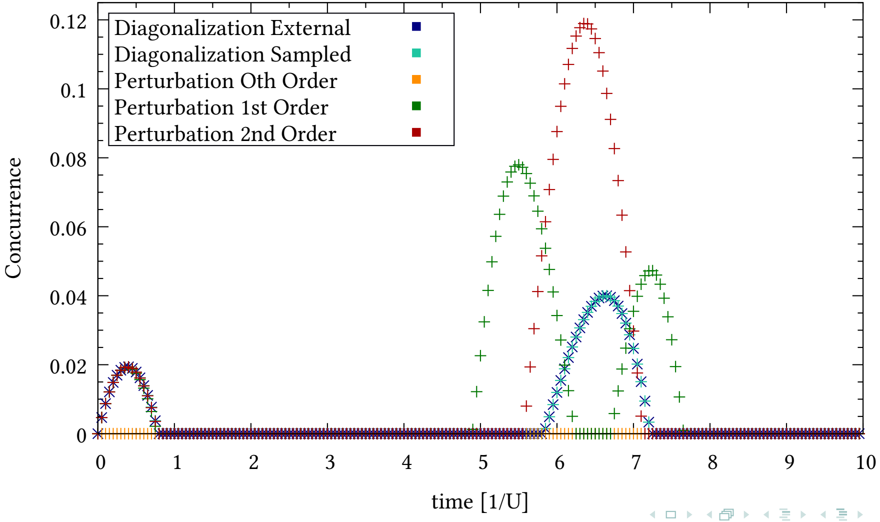
Extra Plots: Overview

Extra slides

- Exact comparison density matrix
- First orders: energy and variance comparison
- J parameter sweep
- Monte Carlo sampling convergence
- Runtime validation
- VCN: parametrization failiure
- VCN: energy conservation can be controlled
- VCN: minimal square reference system
- VCN: larger square reference system in direct comparison
- VCN: system size dependency

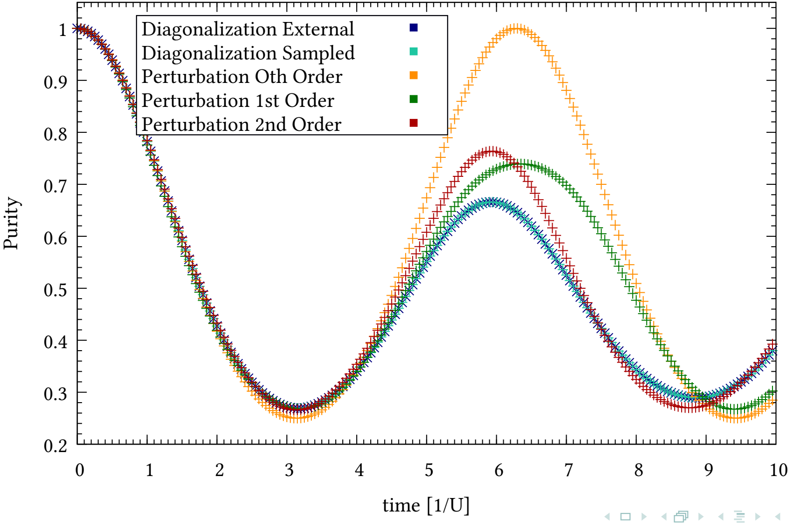
Exact Comparison: Concurrence

Extra slides



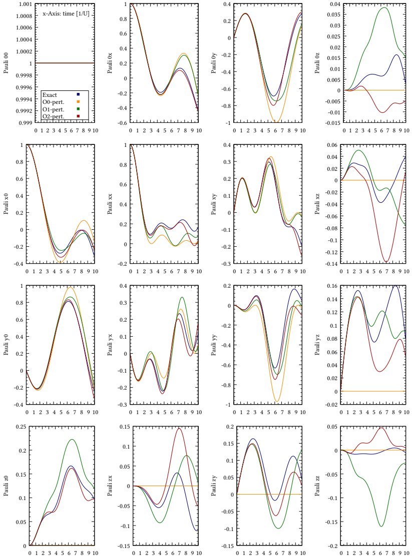
Exact Comparison: Purity

Extra slides



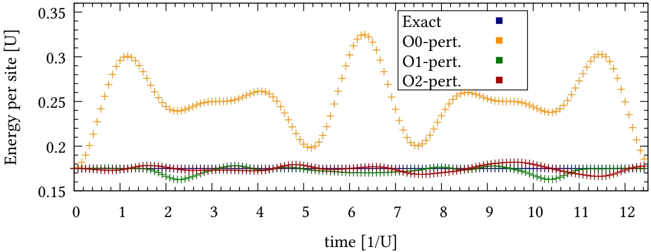
Exact Comparison: Pauli Measurements

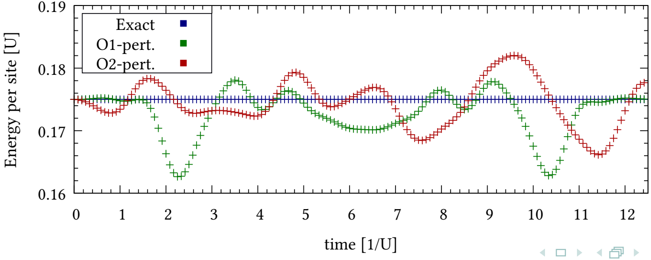
Extra slides



First Orders: Energy Comparison

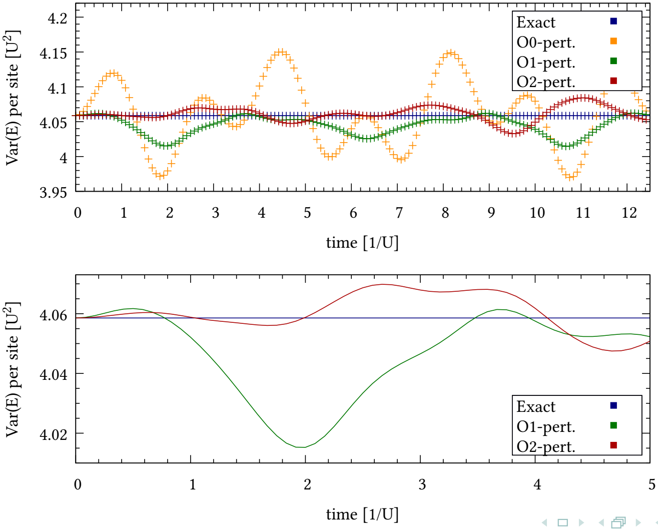
Extra slides





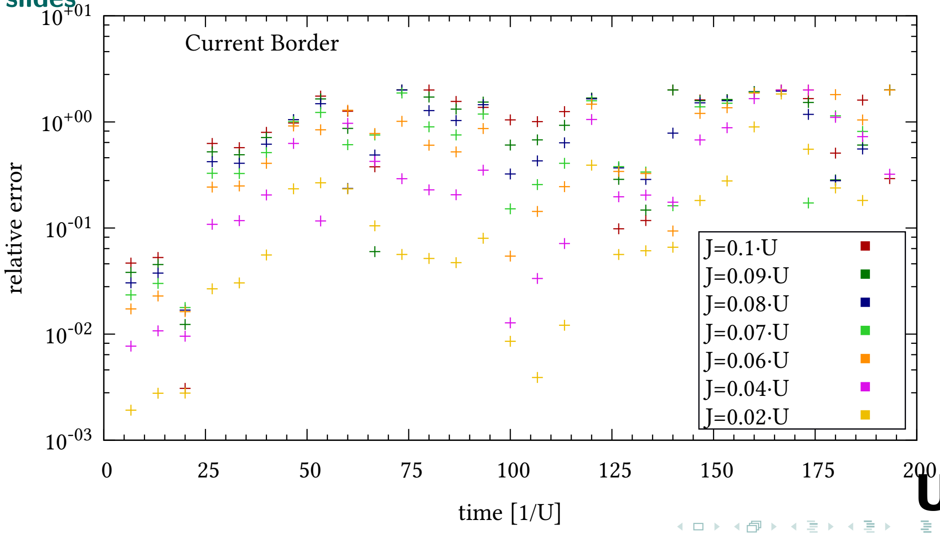
First Orders: Variance Comparison

Extra slides



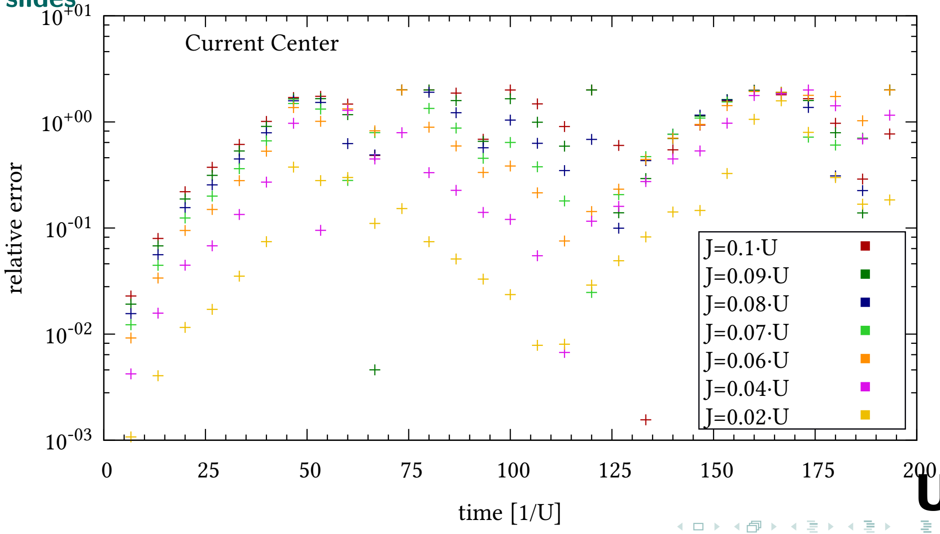
J-Sweep: Current Border

Extra slides



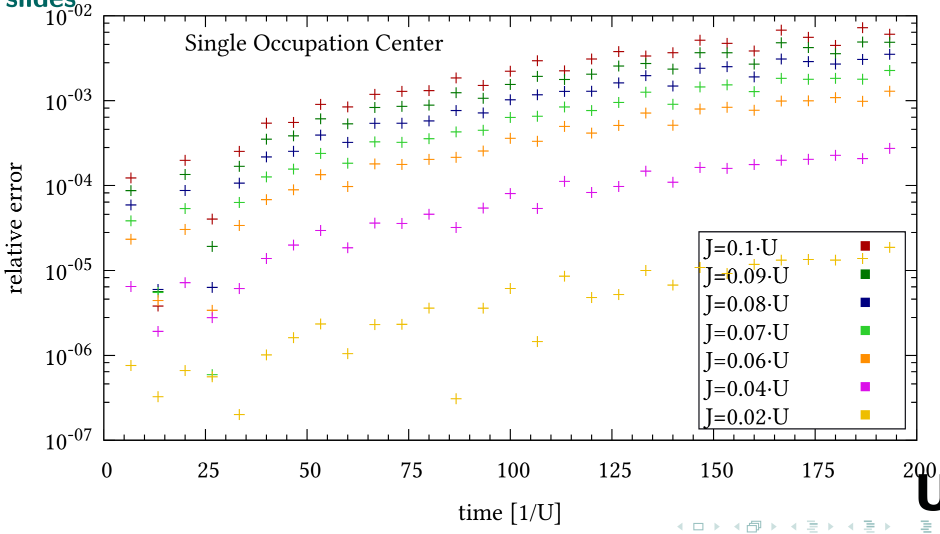
J-Sweep: Current Center

Extra slides



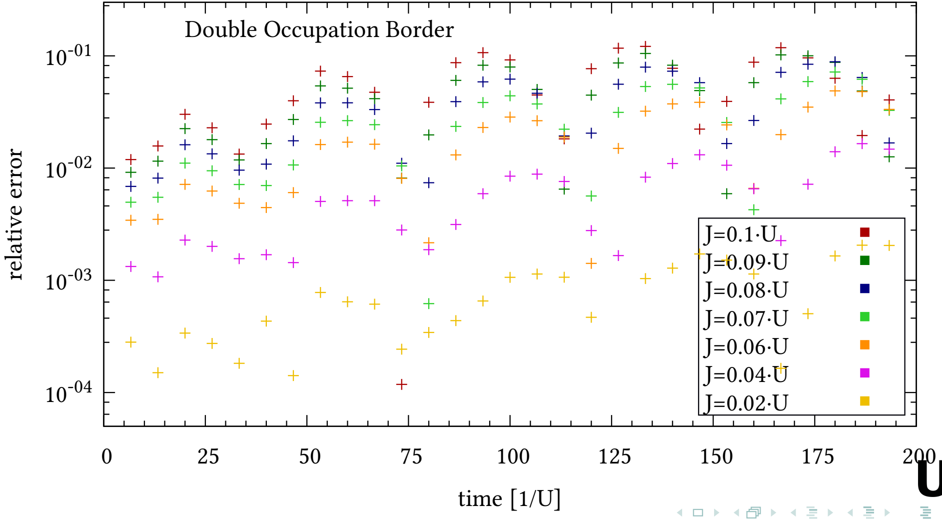
J-Sweep: Single Occupation Center

Extra slides



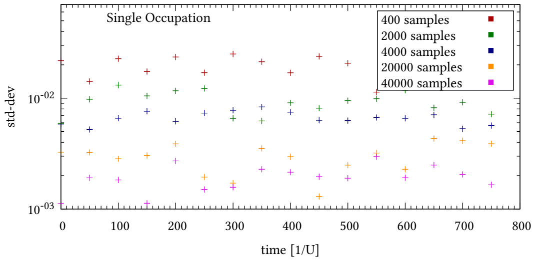
J-Sweep: Double Occupation Border

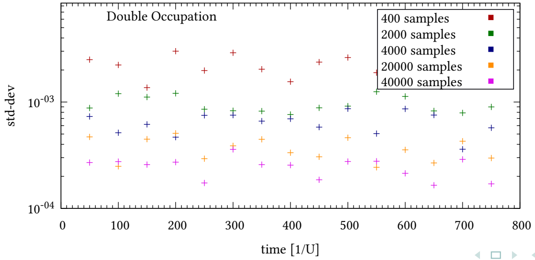
Extra slides



Monte Carlo Convergence: Occupation

Extra slides





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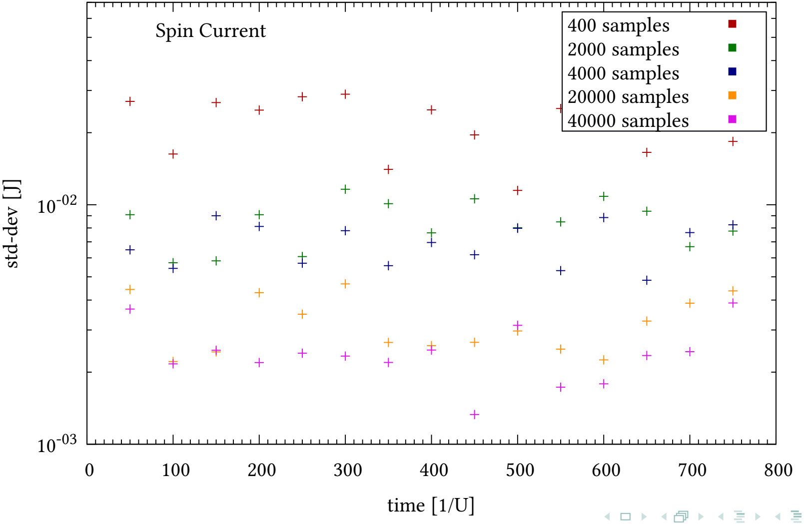
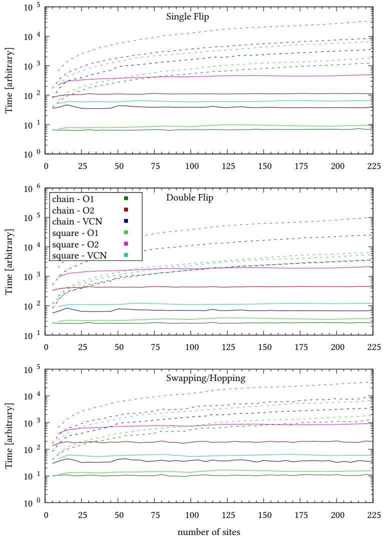


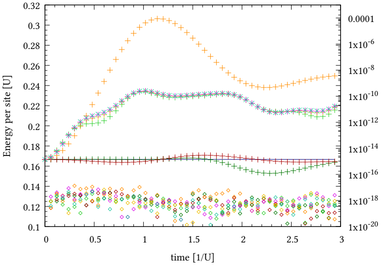
Runtime Validation

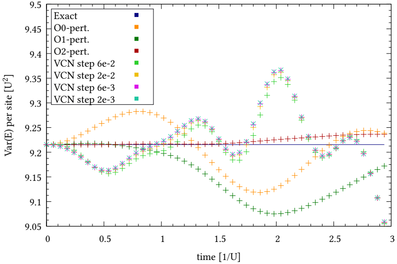
Extra slides



VCN Failiure: Non-Explicit Parametrization

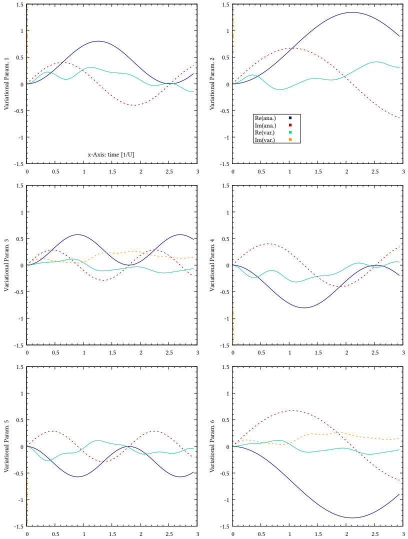
Extra slides





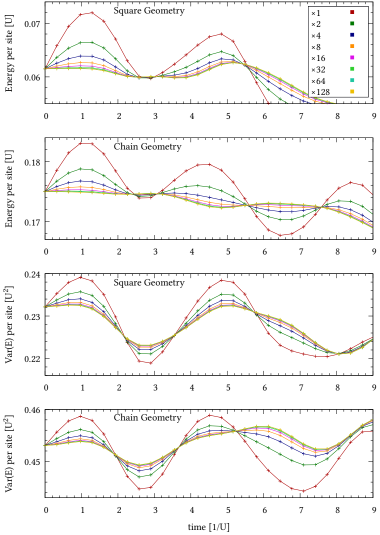
VCN Failiure: Non-Explicit Parametrization

Extra slides



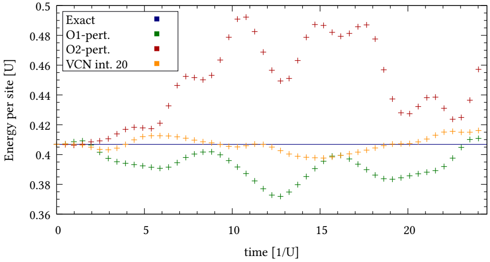
VCN Validation: Energy Conservation Controllable

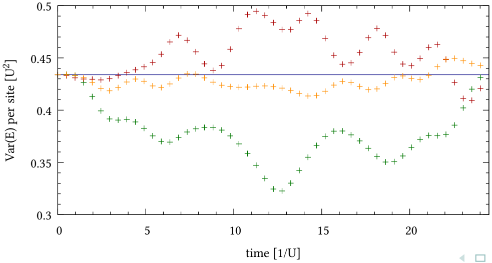
Extra slides



VCN: Small Square System

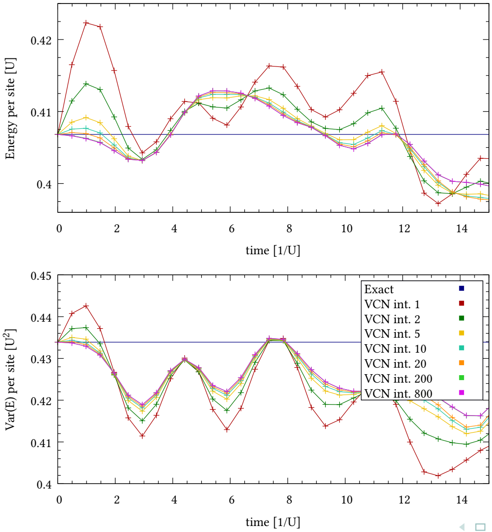
Extra slides





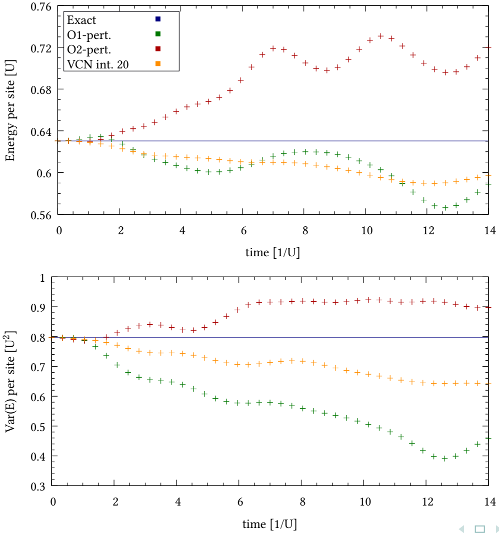
VCN: Small Square System

Extra slides



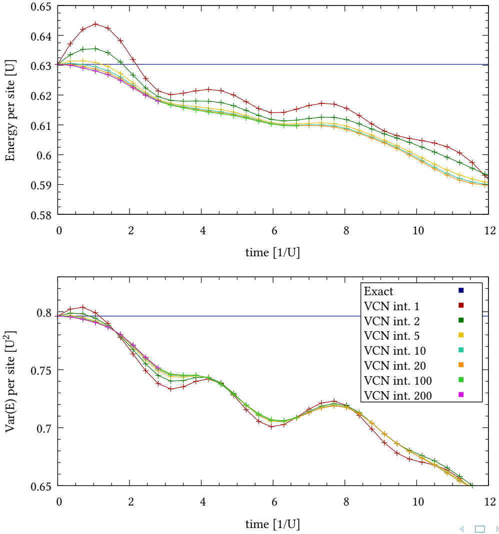
VCN: Bigger Square System

Extra slides



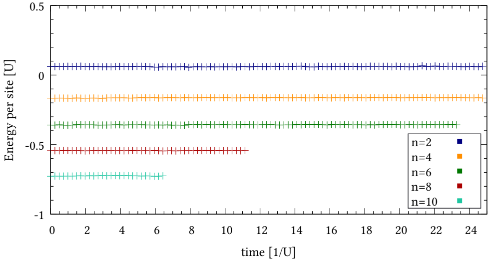
VCN: Bigger Square System

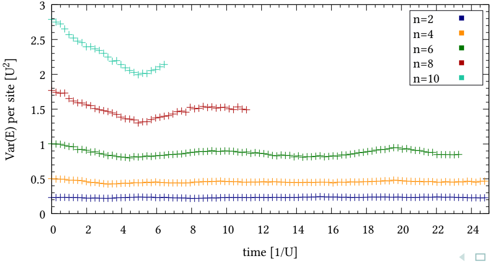
Extra slides



VCN End to End Test: System Size Dependency

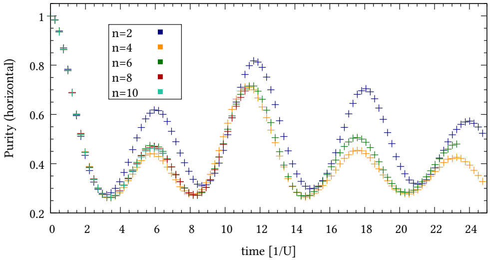
Extra slides

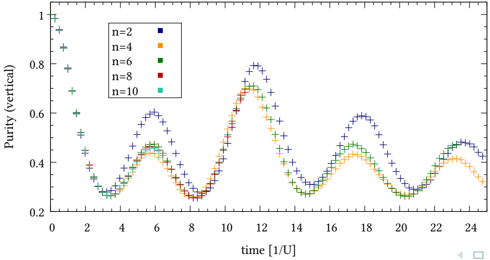




VCN End to End Test: System Size Dependency

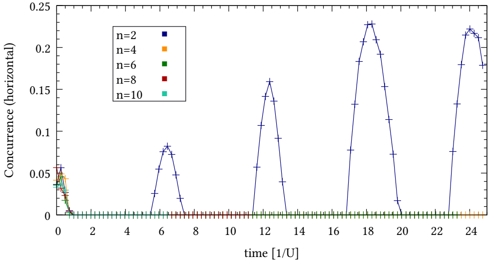
Extra slides

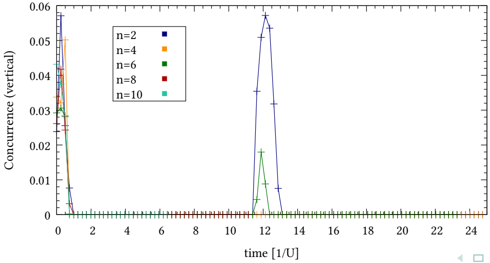




VCN End to End Test: System Size Dependency

Extra slides





VCN End to End Test: System Size Dependency

Extra slides